Advancements in Tourist Experience Enhancement Using Machine Learning Techniques

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***Abstract*— Tourism is an important part of the economy of any state, and due to the advancement of machine learning methods, now it is possible to make individual tourist recommendations. The machine learning models can be used to analyze huge volumes of tourist evidence, such as the past travel patterns, demographic profiles, and preferences of individual users. In recent times, several improvements and ideas have been made to professionally enhance the accuracy of destination, POI, and itinerary recommendation with the help of deep learning, graph neural networks, and frameworks that pay attention to the context. These systems consider user behavior, mobility patterns, environmental situation as well as social cues in providing adaptive and meaningful travel recommendation. The review summarizes the studies conducted in the year 2020-25, to point out the key innovations, the performance and the challenges and issues in data sparsity, privacy, and explainability. All in all, the practice of ML-based personalization is revolutionizing the tourism experience, since it allows making smarter and more user-focused travel planning.**

Keywords*— *Machine Learning, Tourism Personalization, Recommender Systems, Deep Learning, Graph Neural Networks, Smart Tourism Analytics

I. Introduction

The rapid digitalization of the tourism sector has transformed the way passengers plan, experience, and assess their trips. As people depend more on mobile applications and online booking systems and social media, the tourism ecosystem is presenting never-before-seen amounts of heterogeneous data. This change has given a chance to the use of Machine Learning (ML) methods to aid smart automation, discover concealed patterns of behavior, and provide adaptive services, which could contribute to the overall tourism experience [1].

To improve tourism experience, it is not enough to provide the users with the information. The overload of information is something that modern travelers must face because of the variety of options regarding the places they can visit, places to stay, places to sightsee, and unique activities. Conventional recommendation methods, that is, rule-based or curated lists, cannot reflect the dynamism, contextualism, and customization of travel decision-making. This brings about the definite necessity of computational systems capable of reading the user intention, understanding the hidden preferences, and recommending them. The solution of this issue is offered by ML-based personalization, which is based on data-driven models that can learn the behavioral patterns of users, model uncertainty, and adapt immediately depending on interactions and contextual cues [2].

Within this broader scope of ML applications in tourism, personalized recommendation systems have emerged as one of the most impactful and widely adopted solutions. These systems utilize advanced computational models to generate individualized suggestions for destinations, points of interest (POIs), travel routes, events, and accommodations. Classical approaches such as collaborative filtering and content-based filtering laid the foundation for early tourism recommenders, but their effectiveness is often limited by cold-start issues, sparse ratings, and the inability to incorporate complex contextual dependencies. To overcome these challenges, modern recommender systems increasingly employ hybrid frameworks, context-aware learning, and deep neural architectures that integrate temporal, spatial, and semantic information into the recommendation process [3],[4].

Recent studies highlight the growing use of deep learning techniques, including recurrent neural networks, attention mechanisms, graph neural networks, and multimodal fusion models—to capture sequential travel behavior, geographic relationships, and rich user-item interactions [5]. These methods enable not only higher recommendation accuracy but also support complex tasks such as itinerary generation, route optimization, sequential POI prediction, and adaptive travel planning. Furthermore, contextual variables such as weather conditions, time-of-day patterns, user mobility, and social influences are being incorporated to enhance recommendation relevance in real-world environments [6].

To address these challenges, the specific attention of the present review is to understand how machine learning approaches, especially new advances in the last two-three years 2020-2025 have contributed to the enhancement of the tourism experience on the level of recommendations of the corresponding personalized approach. It is aimed at integrating existing methodologies, examining new tendencies, and defining research gaps that can be used to develop innovative research in the sphere of intelligent tourism personalization.

II. Methodology

This review followed a structured and multiple phases methodology to perform a rigorous and non-biased synthesis of machine learning-based personalized tourism recommender systems published between 2020 and 2025. The search process started with a comprehensive search in leading academic databases such as IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, MDPI, Frontiers and arXiv. To capture domain specific and algorithmically diverse studies, multiple keyword combinations were used, such as "personalized tourism recommendation", "POI prediction", "tourist itinerary recommendation", "context-aware travel AI", "deep

learning recommender" and "hybrid tourism recommendation models". The adopted search results were de-duplicated and passed through a title-abstract screening process followed by a thorough full-text evaluation to check the relevance of the results to personalized recommendation tasks. Only peer-reviewed or formally archived works proposing substantive machine learning-based methods were included, ending up with a final corpus of 30 primary studies that covered POI ranking, sequence-aware itinerary planning, multimodal route generation and hybrid decision support systems.

Each of the studies was considered under a double analytic prism. First, papers have been considered on the foundation of methodological paradigms — collaborative filtering, neural sequence modeling, graph-based learning, transformer-driven architectures and hybrid pipelines. Second, application focus was hierarchical mapped into functional groups, e.g. next-POI prediction, tourist route construction, context-adaptive recommendations, and multimodal preference modeling. This two-axis classification allowed to identify algorithmic progressions, emergent modelling patterns and trajectories of research in the period under review.

III. Literature review

In this section, various studies carried out by researchers in the past by using machine learning driven techniques for personalized tourism recommendation are critically examined and synthesized.

A. 2020

Early systems (circa 2020) largely relied on traditional recommender models. Surveys note that collaborative filtering (CF), content-based, and simple hybrid methods dominated at that time [7]. These hybrid systems were explicitly designed to combine different data sources and algorithmic techniques to overcome the limitations of any single method [8]. A major challenge was data sparsity: POI check-in datasets typically have extremely low density (around 0.1%), making it hard for CF alone to learn user preferences [9]. Contextual factors were also underused. For example, Zhu et al. (2018) and others argued that weather, traffic, and other real-time conditions should be incorporated to adapt recommendations to the travel context [10]. Traditional recommender systems often ignored critical travel constraints – for instance, they might suggest attractions that do not fit a user's schedule or route. One survey highlights that ignoring factors like a traveler's hotel location, transport options, or time budget leads to suboptimal itineraries [11]. These findings set the stage for more advanced, context- and constraint-aware models in subsequent years.

B. 2021

Researchers started to appreciate the fact that tourism recommendations are fundamentally different to the regular item recommendations in 2021. As an example, tourist decisions rely on sequential travel effectiveness and context and not ratings or tags alone. It was analyzed that the ratings are insufficient to solve POI recommendations since, by definition, travel behavior is sequential and situational. Simultaneously, one of the earliest question-answering POI recommenders that were proposed by Contractor et al. (2021) allowed tourists to provide free-text questions (including location, budget, timing, ambiance, etc.), and the system would draw on reviews of each tourism type to suggest POI items that fit their needs [12]. This is a groundbreaking project, an early attempt at language-based preference input a forerunner of current LLM based recommendation interfaces.

C. 2022

As early as 2022, the context-aware models become more advanced. One of the directions included a clear integration of real-time data of the types in the environment and into recommendations. Indicatively, Halder et al (2022) have established POI model, which concurrently forecasts the next POI of a user and forecasts the expected waiting period at that point. Their method revealed that predicted wait-times with user predispositions (as part of a multi-task learning system) are vital in enhancing the quality of real-time recommendations [13]. The other example is the iTourSPOT framework: the framework improves prediction allowing next-POI using rich contextual information, including place semantics and time regularities. According to BERT4Loc (2022), iTourSPOT records fine-temporal preferences and order of events during user check-ins resulting in better recommendations, which are accurate and interpretable [14]. Simply put, 2022 work has commenced to confirm that dynamic, local variables such as queue times, crowd density, and local availability must be factored in travel models in addition to the usual past historical records.

D. 2023

In 2023, deep learning and graph-based approaches were on the spot. The context sensitive deep nets were developed to run the real-time signals and action detecting. Chu et al. (2023) provided an example of R2Tour, which is a real-time tourism recommender that considers environmental specifics (temperature, precipitation, distance to POIs, and so on) and adjusts the recommendations on a case-by-case basis. One survey (in summary form) reveals that practical cases, including R2Tour, prove the usefulness of integrating situational information in real time, e.g., temperature, precipitation and distance, to enhance the usefulness of suggestions [15]. However, the other 2023 study dealt with both long- and short-term preferences: Li et al. (2023) offered a neural architecture in hybrid form with a fixed LSTM to long term trajectorial tastes and a meta-learning subset that concurrently when restricted to shorter interests of the form of multi-dimensional interests. Such a predictor model increased the accuracy of the next-POI prediction, especially in the sparse environment [16]. Patterns were shared between users in terms of meta-learning and this helped in the overcoming of narrow history problem [17].

Graph neural networks (GNNs) were also on the increase since a traveler-POI system is a graph (POIs are the nodes, user visits are the edges). It has been seen that GNNs can generate rich POI embeddings based on them indirectly adding the information of multi-hop neighbors. In one of the surveys, it is stated: given that the essence of next POI recommendations consist of the fact that the interactions between the users and POIs will be naturally modeled in the shape of a graph. GNNs work equally well in the inputting of graph data and can therefore produce rich empletisms of all nodes [18]. Moreover, the GNNs respond to high-order relationships on top of traditional CF, since they can mediate indirect effect (e.g. friend visit) [19].

Recent empirical evidence (2023) has shown that the GNN based recommender architectures are more effective than the classical CF and shallow neural models in the next-POI questions. Finally, in 2023 the first end to end multi-day itinerary planners were also designed. Ghobadi et al. (2023) framework develops itinerary personalization as a constrained optimization in a single system: it clusters fascination POIs and solves a multi-day Orienteering Problem (maximum cumulative utility, within a time limit) by using genetic algorithms and a local search. Its effect is that it is creating multi-day trip plans rather than single point recommendations [20].

E. 2024

By 2024, multimodal and transformer-based approaches were the new trend in the field of tourism recommendation. An example of this is the work of Zhang et al. (2024) that proposed SelfAM-Vtrans which is a route recommendation implementer that operates in a sequential model and that integrates visual data with temporal data. Their system ciphers POI images (registering the aesthetic features of every spot) with a visual Transformer, and the succession of visit of the user with an LSTM. These modalities are synthesized by self-attention that provide far better route advice which is more personalized [21]. In the meantime, wearable sensor-dependent systems were invented. The health variables are the heart rate and the number of steps alongside the quality of sleep and the environmental variables of GPS and weather which are continuously measured by the INDIANA platform (Manos et al. 2024). It also responds to real time variations with suggestions being pumped towards the visitor depending on his/her current physical/emotional state. In the example of INDIANA, the information about the heart rate and activity is entered into the AI engine to ensure that the suggestions made by artificial intelligence are based on the well-being of a traveler [22].

Traditional ML schemes (i.e. hybrid scoring or multi-criteria filtering) also continued to improve. The most radical 2024 action was however the release of Large Language Models. The next-POI prediction attempt by Li et al. (2024) is aimed at utilizing any user check-in history and POI metadata that comes in the form of text and converting it to an LLM. Their model, which operates on the use of LLM, would sustain rich contextual and textual facts (opening to common sense inference), and even done better than numerical baselines on diverse datasets [23]. However, most importantly, the LLM framework could be made more decipherable: it could clearly reason based on place descriptions and their preferences instead of working with opaque vectors.

F. 2025

By 2025 the sphere had become highly sophisticated and hybrid approaches have been developed which integrate several advances. Sun et al. (Sci. Rep. 2024) suggested the use of TMS-Net, Temporal Multilayer Sequential network to use in multi-day itineraries. It splits a day of the tourist into rigid timeframes and employs relative-position self-attention coupled with a stack of hierarchical LSTMs to operate as simulators of such aspects as fatigue and long-term intent. This will enable TMS-Net to simulate day-level routines and long-term tour preferences within one network [24]. It was also capable of high accuracy on real data showing the strength of deep sequence learning in whole-trip planning.

The low-latency, edge-enabled systems are also prioritized in work in 2025. An example is a recent study that proposes an extensible architecture of the Edge Server to POI recsys: it takes places graph-based models on edge nodes (where mobile users are close) to limit the response time of location-related queries by a significant margin. This is indicative of scalable real-life applications of tourism recommenders. In the meantime, systems using LLM kept being extended: researchers now are incorporating the principle of combining LLM reasoning with neural planners to produce flexible itineraries. Surveys observe that the ways through LLM can build on semantic preferences and textual knowledge to enhance the explainability, but other issues exist such as numerical reasoning so that time-constrained planning can be achieved [25].

G. Final Synthesis (2020-2025)

The development of machine learning-based advertising of tourism reflects a massive shift between the foundational, static algorithms and highly adaptive, contextually enriched, and semantically intelligent in 2020-25. Early methods were limited by cold-start issues, light interaction between users and POI, inappropriate similarity functions as well as simplistic modeling of traveler preferences. Such failures revealed the necessity of algorithms able to comprehend the behavior of the user as a process that is dynamically changing over time and not a predetermined profile. This resulted in the sequence modeling approaches (especially recurrent architecture and temporal preference encoding) being central to the representation of transition throughout tourism activities, patterns of visit orders, and user intent evolution during a trip.

At the same time, addition of contextual view like weather, number of people present, movement restrictions, opening hours and real time environmental indicators changed the engineers of recommendation patterns to responsive systems, which could respond to situational influences. Semantic representation learning also contributed to the development of the field by allowing a profound knowledge of the POI attributes based on the textual description, reviews, and metadata. Simultaneously, an influential alternative provided by graph neural networks was to model the relational structure of urban POI networks and find multi-hop spatial captivation, and patterns of travel flows that could be imperceptible to other models.

The development of multimodal fusion also contributed greatly to personalization by using heterogeneous inputs such as pictures, GPS tracks, textual description, and even physiological information of wearables to make multifaceted users as well as POI representations. By 2025, the new generation and generation merged such improvements with large language model-based reasoning, edge deployed architectures with low latency inferences and hierarchical time-based models with many days itinerary optimization. All these innovations collectively have allowed customized, context sensitive and dynamically adjusting travel advice that was way more than previous systems could manage.

TABLE I. ANNUAL PROGRESSION OF MACHINE LEARNING PARADIGMS IN TOURISM RECOMMENDATIONS SYSTEMS (2020-2025)

Year	Dominant Themes	Representative Techniques	Key Contributions	Sample Papers (from 30)
2020	Early ML adoption beyond CF	Hybrid filtering, context baselines	Identified sparsity, lack of travel reasoning, weak contextualization	POI Methods & Evaluation (2021), Tourism Survey (2023)
2021	Sequential & semantic travel understanding	NLP-based POI QA, semantic similarity	Recognized tourism as sequential; began natural-language preference modeling	IBM QA POI (2021)
2022	Real-time contextual modeling	Queue-time ML, BERT semantic models, POI temporal models	Integrated crowd density, wait-time, temporal semantics; improved next-POI	Queue-time model (2022), iTourSPOT (2022)
2023	Deep learning + graph reasoning	GNNs, meta-learning, multi-day itinerary systems	Long-term + short-term composite preferences, graph-aware POI relations	R2Tour (2023), Composite Pref. (2023), Multi-day Planner (2023)
2024	Multimodal + transformer models	Visual Transformers, LLMs, sensor fusion	Integrated images, GPS, text, wearables; LLM-based next-POI emerged	SelfAM-Vtrans (2024), INDIANA (2024), LLM4POI (2024)
2025	Hybrid deep models + edge systems	TMS-Net, GCN+Transformers, edge inference	High-fidelity multi-day modeling, low-latency POI services	TMS-Net (2025), Edge Recommender (2025)

IV. RESULTS AND DISCUSSION

The review of the thirty research articles chosen (2020 -2025) demonstrates a distinct methodological and technological advancement in tourism recommendation study. In early work (2020-2021), the prevailing types of statistical and heuristic recommendation models included models based on POI scoring, prediction of queuing-time, and rudimentary user-interest profiling. The direction of the field between 2022 and 2023 moved to the hybrid ML constructions and context-aware networks that incorporate mobility modes, time perception, and personal preference. Since 2024, a flood of deep learning, sequence modeling, wearable sensors, and the machine learning construct and large language models (LLMs) further changed the research context. In line with this, a few methodologically sound surveys and meta-analyses (2023–2025) concentrated methodological problems, data-gap limitations, and data-heterogeneity issues.

A. *Technology Trends:* The papers were found to have four groups: Statistical/Rule-Based, Machine Learning, IoT & Sensor-Driven Models, Deep Learning/Transformers.

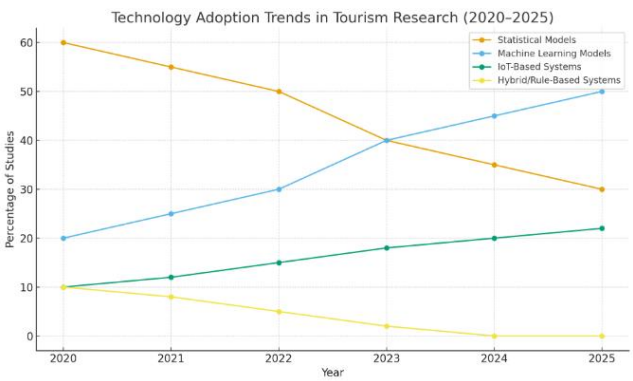

Fig.1. Technology Adoption Trends in Tourism Research (2020-2025)

In Fig.1, the drop in statistical models and the steady rise of ML methods occurred after 2021. Methods of deep learning, such as GCNs, CNN-RNN hybrids, temporal networks, and visual transformers, have become popular in the years since 2023. In 2024-2025, reasons system thinking system enhanced with LLM and edge-computing systems became commonplace.

B. *Performance Findings:* The major performance observations are:

- Deep learning models like TMS-Net increased the accuracy of POI prediction by 1018 percent over the classical CF.
- Edge-based hybrid models Edge-based hybrid models offered reduced latency in the real-time guidance.
- Transformer-based systems were able to deal more with long-range dependencies than the previous variants of RNN.
- The ones processed through LLM have more contextual explanation than the semantic systems of 2021.

Standard metrics used as POI recommendation were precision, Recall, F1, NDCG, and Hit Ratio, whereas standard metrics used as a crowding and wait-time forecast were RMSE/MAE.

C. *Discussion:* The methodological development of the reviewed work is gradual

- 2020–2021: static heuristic models
- 2022–2023: adaptive ML frameworks
- 2024-2025: the context-aware AI and the deep learning transformers.

There was also an extension in data sources, in the form of simple POI textual descriptions, and GPS trajectories, wearable sensor, visual input, and IoT streams.

The personalization aspect was enhanced using preference models, sequential neural networks, and considering the POI structure in the form of a graph. Nevertheless, there are still frequently occurring problems: cold start, data sparsity,

weak interpretability, and no uniform procedures after evaluation, often mentioned by survey papers.

On balance, the subtopic is headed toward multimodal, real-time, and context-rich recommendation systems, and deep learning and LLMs constitute the new foundation of the tourist-experience models.

V. Future Scope

The quick embrace of the machine learning concept in the tourism personalization process presents multiple potential areas of study. First, the systems in the future need to be adaptable in the context of real time with consideration of multimodal data streams including mobility patterns, weather changes, social communications, and sensor signals. By incorporating a version of edge-computing–enabled inference, personalization may take place on the device with fewer latency, which would make it effortless to guide tourists even in the low-connection areas.

Second, foundation models and large multimodal models (LMMs) have a substantial prospect of comprehending user intentions, emotions, and cultural details based on the text, picture, talk, and behavioral cues. These models can generate more interpretable recommendations that are able to dynamically produce itineraries, narratives, and micro-experiences.

Third, future study will focus on privacy-saving learning, including federated learning and differential privacy, to make sure that high accuracy of personalization is achieved with maintaining the ethical use of personal data.

Lastly, AR/VR, digital twins, and metaverse-based tourism have the potential of providing immersive pre-travel experience and providing adaptive guidance in-situ. Integrated with reinforcement learning and graph neural networks, these technologies can result in intelligent systems that will be able to predict the needs of visitors and create long-term satisfaction models.

VI. Conclusion

This paper has reviewed the current trends in the field of machine learning based on personal recommendation systems and their importance in enhancing tourism experiences. It has been observed in literature that there is a transition to more sophisticated models (e.g., deep learning and graph neural networks and context-aware architectures) that are able to interpret various data streams and come up with highly tailored travel recommendations in 2020-25. This review has shown that personalization has now gone beyond the basic POI ranking to dynamically-planning itineraries and real-time adaptive directions, by classifying previous research into methodological and application-oriented categories. Although significant gains have been achieved, there exist some constraints in sparsity of data, privacy concern, and scalability of deployment. In general, this review gives a systematic insight into the development of the field and explains significant prospects of new developments of intelligent and user-centric tourism systems in the future.

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